CMAN – Workload Accounting by Hand.

Network : [784 → 256 → 128 → 10], batch size = 1, FP32, no biases.

a) <u>per – layer MACs</u>

for a fully connected layer:

$$\boxed{\text{MACs} = \text{input neurons} \times \text{output neurons}}$$

Layer 1 : 784 → 256

$$MACs_1 = 784 \times 256 = 200{,}704$$

Layer 2 : 256 → 128

$$MACs_2 = 256 \times 128 = 32768$$

Layer 3 : 128 → 10

$$MACs_3 = 128 \times 10 = 1280$$

b) <u>Total MACs</u>

$$\text{Total MACs} = MACs_1 + MACs_2 + MACs_3$$

$$= 200{,}704 + 32768 + 1280$$

$$= 2{,}34{,}752$$

c) <u>Total trainable parameters (weights only)</u>

for a fully connected layer

$$\boxed{\text{Weights} = \text{input neurons} \times \text{output neurons}}$$

layer 1 :

$$W_1 = 784 \times 256 = 200704$$

layer 2 :

$$W_2 = 256 \times 128 = 32768$$

layer 3 :

$$W_3 = 128 \times 10 = 1280$$

Total Weights:

$$W_{total} = W_1 + W_2 + W_3$$

$$= 200704 + 32768 + 1280$$

$$W_{total} = 234,752$$

d) <u>Total weight memory in bytes (FP32)</u>

Each FP32 weight = 4 bytes

$$\text{weight bytes} = W_{total} \times 4$$

$$= 2,34,752 \times 4$$

939,008 bytes

e) <u>Total Activation memory in bytes</u>

we store all activations at once:

- input : 784
- Hidden 1 output : 256
- Hidden 2 output : 128
- output : 10

Total Activations:

$$A_{total} = 784 + 256 + 128 + 10 = 1178$$

each Activation is FP32 = 4 bytes:

$$\text{Activation bytes} = A_{total} \times 4$$

$$= 1178 \times 4$$

$$= 4712 \text{ bytes}$$

f) <u>Arithmetic intensity (FLOP/byte)</u>

$$\text{Arithmetic intensity} = \frac{2 \times \text{total MACs}}{\text{weight bytes} + \text{Activation byte}}$$

$$= \frac{2 \times 234{,}752}{939{,}008 + 4712}$$

$$= 0.4975 \text{ FLOP/byte}$$

$$\boxed{\text{Arithmetic intensity} = 0.4975 \text{ Flop/byte}}$$